

MAT282 · INTERMEDIATE STATISTICS · FALL 2026 · VERSION B

Answer key

Version B

Marking guide and worked solutions · 40 points · **instructor copy: do not distribute before the exam**

How to mark. Sections 1 and 5 and the explanation parts of the others reward **sensible reasoning that is defended with a reason or a number**; there is no single right model. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Students interpret coefficients but are **not yet** asked whether coefficients are significant. In Section 4 a t-test is **significant if $p < 0.05$** (two-tailed, T.TEST type 2). Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: temperature (up: iced drinks sell in the heat), weekend (up: more customers), beach event (up: crowds nearby), planned price (down: dearer coffee sells less), sunshine (up, overlaps with temperature).	3	1 per input: direction and a coherent reason.
1b	LO5 · Con	Item 6 , iced coffee revenue calculated after closing: it is not known before opening and it contains the outcome (price $\times$ cups sold), so it cannot be used to forecast cups.	2	1 identifies item 6 + 1 the problem (not known in advance and contains the outcome).
1c	LO3, LO5, LO6 · Con	Not both. With $r = 0.78$ they carry mostly the same information, so the model cannot tell their effects apart ("holding sunshine constant while temperature changes" is unrealistic): the coefficients become unreliable. Keep temperature ($r = 0.68$ with sales vs 0.53 ; a more direct measure of heat).	2	1 overlap and why it is a problem + 1 a choice with a reason.
1d	LO6 · Con	Hold back some days, fit the model on the rest, predict the held-back days, then calculate their residuals and MAE.	1	1: a hold-out check with residuals/MAE (names the idea of unseen days).
2a	LO5, LO6 · Com	Sales = $100.04 + 3.45 \times \text{temperature} + 18.09 \times \text{weekend} - 21.58 \times \text{price}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; the table shows Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 correct numbers matched to the right variables + 1 correct equation form.
2b	LO6, LO7 · Con	(i) 21.58 is the change per €1 : a €0.20 rise lowers predicted sales by $0.20 \times 21.58 = 4.3$, about 4 cups, holding temperature and weekend fixed. (ii) weekend = 0 only means there is no weekend adjustment: the model still predicts sales from the intercept, temperature and price; weekends add about 18 cups. (iii) A typical forecast misses by about 12 cups ($SE - Y$); some by more, some by less.	3	1 each: corrected statement with units and conditions.
2c	LO3, LO6 · Cal + Con	(a) $0.20 \times (-21.58) = -4.32$, about -4 cups . (b) 4 °C cooler: $-4 \times 3.45 = -13.8$, about -14 cups . (c) The coefficients are per €1 and per 1 °C, so their sizes depend on the units. A €0.20 price change moves sales less (-4.3) than a 4 °C change (-13.8): compare realistic changes, not raw coefficients.	3	1 + 1 + 1.
2d	LO5, LO6 · Cal	(a) $\hat{y} = 100.04 + 3.45(24) + 18.09(0) - 21.58(3.3) = \mathbf{111.6}$ cups. (b) Residual = $118 - 111.6 = \mathbf{+6.4}$. (c) residual = 6.4 is smaller than $SE - Y = 12.25$: a good forecast: the miss is smaller than the typical error.	4	2 prediction (1 substitution, 1 value) + 1 residual (actual - predicted) + 1 comparison with $SE - Y$.
3a-i	LO3, LO6 · Cal	Filled table: Tue actual 100, predicted 108, residual -8; Sun actual 130, predicted 120, residual +10; Thu actual 96, predicted 93, residual +3.	1	1 all blanks correct (follow-through on a single slip).
3a-ii	LO3, LO6 · Cal	MAE = $(8 + 10 + 3) \div 3 = \mathbf{7}$ cups.	1	1 correct (absolute values used).
3a-iii	LO6 · Con	Model Z has the lower MAE (5 vs 7). But every residual has the same sign: Z under-predicts every day by 5 (systematic bias). A simple correction (add 5) would remove it. Model X's errors are larger but not one-sided.	2	1 lower MAE with numbers + 1 identical signs = bias.

3b	LO3, LO4, LO6 · Con	(i) Yes: Day 8 is the hottest day with the highest sales and lies close to the line of the other days (Figure 1). (ii) Slope 4.41 with it vs 4.25 without; R^2 0.98 vs 0.97: almost no change. (iii) Keep it : it is extreme in size but follows the pattern, and deleting it would throw away information about hot days (and the model would extrapolate to them).	4	1 follows the pattern + 2 impact (small change in slope and R^2) + 1 keep with a reason.
4a	LO3 · Cal	Means 100 and 114; $\Sigma(x - \bar{x})^2 = 160$ and 320; $s_1 = \sqrt{(160 \div 4)} = 6.32$, $s_2 = \sqrt{(320 \div 4)} = 8.94$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.
4b	LO3 · Cal	$s_p = \sqrt{[(4 \times 40 + 4 \times 80) \div 8]} = \sqrt{60} = 7.75$; $t = (100 - 114) \div [7.75 \times \sqrt{(1/5 + 1/5)}] = -14 \div 4.90 = -2.86$.	2	1 s_p + 1 t (follow-through on s; sign may be reported either way if direction is stated).
4c	LO1, LO3, LO6 · Con	(i) H_0 : the price cut made no difference to mean daily iced coffee sales (mean before = mean after). (ii) $p = 0.021 < 0.05$, so the difference is statistically significant : reject H_0 . (iii) If the price had no effect, a difference this large or larger would turn up in only about 2% of samples like these. It is not the probability that H_0 is true and it does not give the size of the effect.	3	1 null hypothesis + 1 decision using p + 1 correct plain-words meaning.
4d	LO6 · Con	Warmer weather also raises iced coffee sales, so the rise cannot be credited to the price cut alone (the weather is mixed in with the price). To separate them, put price and temperature in the same LINEST model.	1	1 a relevant reason/idea.
5	LO7 · Con	(i) The inputs recommended (e.g. temperature and weekend push sales up and a higher price pulls them down (not sunshine, which overlaps with temperature)) with the typical forecast error (about 12 cups). (ii) The price test: iced coffee sales were higher after the price cut (about 14 cups, $p = 0.021$). (iii) A caution (small samples, other changes, association not proof, forecasts can still miss by the typical error).	3	1 inputs + main LINEST finding in plain words + 1 price test result in plain words + 1 a genuine caution.

2 · Worked tables

Section 4 worksheet (Table 6).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	100	0	0	110	-4	16
2	108	+8	64	118	+4	16
3	92	-8	64	102	-12	144
4	104	+4	16	114	0	0
5	96	-4	16	126	+12	144
Sum	500	0	160	570	0	320
Mean $\bar{x}$	100			114		
s	6.32			8.94		

Check against Sheets: =T.TEST gives $p = 0.0212$ ($df = 8$, $|t| = 2.858$, two-tailed).

3 · Where students go wrong

- **Reading the LINEST table backwards** (2a): Col A is the **last** X column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Per customer” and “causes”** (2b): the coefficients describe the **daily total** and an **association**.
- **Reading R^2 as “percent correct”** (2b): R^2 is the share of variation explained.
- **Using absolute values** (3a-ii): MAE averages the sizes of the errors, so signs must not cancel.
- **Reading p as the probability that H_0 is true** (4c). Credit “if there were no real effect, a result this extreme would be rare”.